

Email: jaredhwa@cs.washington.edu

Website: jared-hwang.github.io

LinkedIn: linkedin.com/in/jared-hwang/

Phone: +1 (929) 399-6236

Education

- 09/2024 - Present **University of Washington**, Seattle, WA
Ph.D. Computer Science & Engineering
Advisor: Prof. Jon E. Froehlich
- 06/2021 - 08/2023 **University of Southern California**, Los Angeles, CA
M.S. Computer Science
GPA: 4.00, M.S. Honors Program
- 08/2016 - 05/2020 **Tufts University**, Medford, MA
B.S. Physics, B.S. Computer Science
GPA: 3.94, summa cum laude, Sigma Pi Sigma Honor Society

Conference Publications

- [C2] **“Where can I Park?” Understanding Human Perspectives and Scalably Detecting Disability Parking from Aerial Imagery** [\[pdf\]](#)
Jared Hwang, Chu Li, Hanbyul Kang, Maryam Hosseini, Jon E. Froehlich.
In Proceedings of ASSETS'25
- [C1] **Digital Typhoon: Long-term Satellite Image Dataset for the Spatio-Temporal Modeling of Tropical Cyclones** [\[pdf\]](#) [\[Web\]](#) [\[Code\]](#)
Asanobu Kitamoto, Jared Hwang, Bastien Vuillod, Lucas Gautier, Yingtao Tian, Tarin Clanuwat.
In Proceedings of NeurIPS'23

Workshops, Posters, and Extended Abstracts

- [W2] **RAMPNET: A Two-Stage Pipeline for Bootstrapping Curb Ramp Detection in Streetscape Images from Open Government Metadata** [\[pdf\]](#)
John S. O'Meara, Jared Hwang, Zeyu Wang, Michael Saugstad, Jon E. Froehlich
ICCV'25 Workshop on Vision Foundation Models and Generative AI for Accessibility
- [W1] **“Does the cafe entrance look accessible? Where is the door?” Towards Geospatial AI Agents for Visual Inquiries** [\[pdf\]](#)
Jon E. Froehlich, Jared Hwang, Zeyu Wang, John S. O'Meara, Xia Su, William Huang, Yang Zhang, Alex Fiannaca, Philip Nelson, Shaun Kane
ICCV'25 Workshop on Vision Foundation Models and Generative AI for Accessibility

Research Experience

- 09/2024 - Present **UW Makeability Lab**, University of Washington, Seattle, WA, USA
Advisor: Prof. Jon E. Froehlich
- Created *BikeButler*, a bike-routing platform that allows users to generate, view, customize, and save bicycle routes based on fine-grained preferences on eight 'bikeability' features. Conducted 16 user-study interviews.
 - Developed and evaluated *AccessParkCV*, a CV pipeline that automatically detects and measures the width of disability parking spaces from aerial imagery. Created and released an 11,762 object dataset, and conducted 11 needfinding interviews [\[C2\]](#)
- 01/2023 - 08/2023 **NII Kitamoto Lab**, National Institute of Informatics (NII), Tokyo, Japan
Advisor: Prof. Asanobu Kitamoto
Designed and built Pyphoon2, a dataloader for the Digital Typhoon Dataset. Designed and ran ML benchmark experiments, including classification, regression, and forecasting of typhoon intensity. Work published to NeuRIPS 2023 Datasets and Benchmarks. [\[C1\]](#)
- 05/2020 - 09/2021 **Tufts Neutrino Group**, Tufts University, Medford, MA, USA
Advisor: Prof. Taritree Wongjirad
Designed and ran ML experiments on neutrino detector image datasets. Created autoencoder based image compression model, producing 80% smaller file sizes compared to JPEG. Prototyped spatial loss algorithm for detector image segmentation.
- 06/2018 - 03/2020 **MIT Laboratory for Nuclear Science**, MIT, Cambridge, MA, USA
Advisor: Prof. Janet Conrad and Dr. Daniel Winklehner
Developed a particle accelerator simulation code for radio frequency quadrupoles (RFQ) in the IsoDAR experiment. Ran simulations with greater accuracy and 20% faster than competing solutions. Included visualization and parallel computing capability. [\[Code\]](#)
- 09/2017 - 05/2018 **Tufts Sajina Lab**, Tufts University, Medford, MA
Advisor: Prof. Anna Sajina
Performed statistical analysis and crossmatching of two astronomical satellite datasets.

Projects

- 11/2021 - 12/2021 **Simulation Inference on Urban Traffic Data**
University of Southern California, Los Angeles, CA
Applied Bayesian conditional density estimation simulation inference to traffic simulations of the Seattle area. [\[Web\]](#)

Industry Experience

- 10/2023 - 10/2024 **Software Development Engineer**
Amazon, Seattle, WA
Maintained massive pipelines and databases for customer behavior and purchase data.
- 05/2022 - 08/2022 **Software Development Engineer Internship**
Amazon, Seattle, WA
Integrated Apache Airflow workflow orchestrator for ML pipelines.

Invited Talks

- 10/2025 **Pacific Northwest Transportation Consortium (PacTrans) '25**
BikeButler: Creating and Previewing Personalized, Context-Sensitive Bicycle Routes
- 06/2023 **Fujitsu & YNU Typhoon Research Lab Joint Research Workshop**
Machine Learning for the Digital Typhoon Dataset

Service

Paper Reviews

- ICCV'25 Workshop on Vision Foundation Models and Generative AI for Accessibility

Teaching Assistant

- UW CSE160 Data Programming (Wi'25, Fa'25)
- USC CSCI576 Multimedia Systems Design (Fa'22)
- Tufts COMP105 Programming Languages (Wi'20)
- Tufts COMP40: Machine Structure and Assembly Language Programming (Fa'18, Wi'20)

Honors

- 05/2020 **N. Hobbs Knight Prize Scholarship**
Tufts University, Medford, MA
For demonstrating outstanding ability in theoretical and experimental physics

Technical Skills

Programming

Python, C++, C, Java, JavaScript, SQL, R

Machine Learning & Development

PyTorch, Tensorflow, PyTorch Lightning, Docker, Slurm, Apptainer

Language

Conversational Japanese, Conversational Mandarin